

# Priyanjali Patel

Data Scientist | GenAI, LLM & RAG | Ph.D. in Astrophysics

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## Professional Summary

**Data Scientist** with hands-on experience building **GenAI applications** — Retrieval-Augmented Generation (RAG) pipelines, multi-agent LLM workflows, and LLM-powered multimodal/clinical NLP systems — alongside deep learning work in computer vision (CNN/U-Net segmentation) and sequence modelling (DNN/LSTM), using **OpenAI GPT, Gemini, and GROQ-hosted LLMs** with **LangChain and LangGraph**. PhD-trained in statistical analysis, data modelling, and scientific computing, with experience working with multimodal data including text, tabular, time-series, and image data. Experienced integrating LLMs with REST APIs, SQL databases, and FastAPI/Streamlit applications; skilled in **prompt engineering, vector search, and multi-agent orchestration**, translating complex model outputs into business-ready outcomes for cross-functional, multicultural stakeholders.

## Work Experience

### MEDITECH INNOVATIONS

08/2025 – Present

*Data Science Intern · Remote, Noida West, India*

[Data: Multimodal (medical images: MRI, CT, X-ray, ECG, ultrasound, histopathology) | Text (clinical NLP) | Tabular (health records) | Time-series (clinical sequences)]

- Built **Medical Visual Question Answering (VQA)** system using **Gemini multi-modal model (3.6 Flash)** for AI-assisted interpretation across seven clinical imaging modalities; designed prompt-engineered extraction templates that pull diagnoses, symptoms, and treatment plans as structured tables from unstructured clinical notes.
- Built **ChatRIA**, a **Retrieval-Augmented Generation (RAG)** assistant for pharmaceutical regulatory Q&A: parsed and chunked FDA drug-label PDFs using **LangChain**, embedded content with **OpenAI embeddings** into a **FAISS vector store**, and used **MMR retrieval** with **GPT-4o** to generate structured, source-cited answers; deployed via a **Streamlit** interface.
- Applied ML/DL models (Logistic Regression, DNN, LSTM on **sequential clinical time-series**) for predictive insights; exposed LLM and analytical outputs via **Streamlit dashboards integrated with REST APIs** for real-time clinical decision support.

### MERCOR

05/2026 – 06/2026

*Computational PhD (Contractor) · Freelance, Remote (Physics)*

- Contracted via Mercor as a **computational-physics contributor**, working on AI-related project work for a Mercor client.

### Collaborative Project – ML-based Fraud Detection

11/2025 – 01/2026

[Data: Structured tabular (imbalanced financial transaction records)]

- Analysed **imbalanced tabular financial transaction data** to build a fraud detection classifier (**XGBoost**); translated precision-recall trade-offs into actionable threshold recommendations aligned with business risk tolerance.

## Personal Projects

### Glacier Segmentation with U-Net

Deep Learning / Computer Vision

[Data: Synthetic multi-band satellite imagery]

- Extended the **U-Net segmentation** work from Omdena into a standalone, open-sourced **PyTorch** pipeline — patch-based dataset generation, multi-band preprocessing, and **Dice-loss training** — to practise the full deep learning workflow independently, end to end.

[github.com/priyanjalipatel/glacier-segmentation-unet](https://github.com/priyanjalipatel/glacier-segmentation-unet)

## Technical Skills

### GENAI, LLM & AGENTIC AI

- LLM APIs: OpenAI GPT, Gemini Pro Vision, GROQ
- LangChain, LangGraph, CrewAI
- Retrieval-Augmented Generation (RAG)
- Vector Search & Embeddings (FAISS)
- Prompt Engineering, Multi-Agent Workflows
- Tool Calling, LLM-based NLP Pipelines
- Hugging Face, Transformer Architectures (built GPT from scratch)

### Machine Learning & Deep Learning

- CNNs, U-Net (Computer Vision / Semantic Segmentation)
- DNN, LSTM (Sequence Modelling)
- XGBoost, Random Forest, Gradient Boosting
- Isolation Forest, Anomaly Detection
- Bootstrap Resampling & Uncertainty Quantification

### Languages & Backend

- Python (Pandas, NumPy, scikit-learn, PyTorch)
- SQL (Intermediate), R (basic)
- FastAPI; REST API Integration
- PySpark, Databricks

### Cloud & MLOps

- AWS (S3, EC2, SageMaker)
- Docker
- Git, Agile Methodology

## Work Experience (cont'd)

### OMDNA

09/2023 – 08/2025

*Machine Learning Engineer (Volunteer) · Remote, Palo Alto, USA*

[Data: Image (multi-band Landsat satellite imagery) | Text (financial documents, PDFs, queries) | Structured relational (SQL)]

- Designed, trained, and evaluated a **U-Net convolutional neural network (PyTorch)** for pixel-wise glacier segmentation on multi-band Landsat satellite imagery — building the deep learning pipeline end-to-end (patch-based dataset generation, multi-band pre-processing, **Dice loss** to handle severe foreground/background class imbalance) and reaching a **validation Dice coefficient of ~0.82**.
- Designed and built a **multi-agent Retrieval-Augmented Generation (RAG) + SQL chatbot** using **GROQ-hosted LLMs** for an AI banking platform, orchestrating retrieval over unstructured financial text with **tool-calling** into structured transactional data; led the chatbot team's **agent-workflow design**, contributed to Fraud Model integration, and delivered data-driven insights through automated analytics workflows.

### DATA GLACIER

04/2025 – 05/2025

*Data Scientist Intern · Remote, Prayagraj, India*

[Data: Structured tabular (customer banking records — demographic & behavioural features)]

- Built classification models (Logistic Regression, Random Forest, Gradient Boosting) predicting term deposit subscriptions for a BFSI marketing campaign; achieved **ROC-AUC of 0.92** via **SMOTE class balancing** and validated outputs through confusion-matrix / classification-report analysis.

### VIRIDIEN (formerly CCG) + S2DS Fellowship

02/2024 – 03/2024

*Data Engineer · Remote, London, UK*

[Data: Text (unstructured source code & technical documentation for LLM-based refactoring)]

- Selected for the competitive S2DS Fellowship; built scalable **ETL pipelines** to extract and transform unstructured code and technical documentation using **LangChain and OpenAI GPT APIs** for **LLM-based automated code refactoring** in an Agile team for a global HPC leader.

### UNIVERSIDAD DE CHILE

03/2019 – 12/2024

*Doctoral Researcher · Santiago, Chile*

[Data: Time-series (large-scale multi-wavelength photometric survey, ~7,000 astronomical objects)]

- Calculated quasar optical variability using **C code (Mexican Hat code)**, identifying and fixing a critical error to enhance short-timescale variability detection.
- Conducted analysis of quasar optical variability using Python and advanced statistical techniques: retrieved and processed large-scale time-series data (light curves) for ~7,000 quasars via APIs from the **Zwicky Transient Facility (ZTF)** survey; applied statistical modelling (**bootstrapping with 1,000 samples, regression analysis**).
- Applied **Bayesian inference / Markov Chain Monte Carlo (MCMC)** techniques for parameter estimation, uncertainty quantification, and extracting insights from complex, noisy time-series datasets.
- Developed Python-based **data analysis pipelines** for the **ALeRCE ML collaboration**, improving processing efficiency by **15%**; performed quality filtering of time-series data for ML-ready astronomical datasets.
- Contributed as a co-author to two scientific papers, involving text review and preliminary investigations of light curves.
- Delivered presentations on research findings at scientific conferences, both locally and internationally.

### UNIVERSIDAD DE CHILE

03/2020 – 01/2023

*Teaching Assistant · Santiago, Chile*

- Graded astronomy exams for five semesters and developed well-structured exams for one semester.

## Publications

P. Patel et al., *Probing the rest-frame wavelength dependence of quasar variability*, A&A, 695, A162 (2025).

P. Arévalo et al., *Universal power spectrum of quasars in optical wavelengths*, A&A, 684, A133 (2024).

P. Arévalo et al., *Optical variability in quasars: scalings with BH mass and Eddington ratio*, MNRAS, 526, 6078 (2023).

## DATA ANALYSIS & STATISTICS

- EDA, Data Cleaning & Wrangling
- Statistical Modelling
- Hypothesis Testing, A/B Testing
- Regression & Time-Series Analysis
- Bayesian Modelling, MCMC
- On-Chain / Web3 Data (Dune Analytics)

### Visualization & Reporting

- matplotlib, seaborn, Plotly
- Streamlit (interactive AI apps)
- Business Dashboard & KPI Reporting

### Soft Skills

- Stakeholder Communication
- Cross-functional Collaboration
- Data-Driven Decision Making

## Education

### Ph.D. in Astrophysics

Universidad de Chile

Santiago, Chile | 2019–2024

ANID National Fellowship Recipient

### M.Sc. in Physics

Pt. Ravishankar Shukla Univ.

Raipur, India | 2015–2017

### B.Sc. Physics, CS & Math

St. Thomas College

Bhilai, India | 2012–2015

## Awards

### ANID PhD Fellowship

National Research & Development Agency, Chile | 2019–2024

## Languages

English (Fluent)

Hindi (Native)

Spanish (Intermediate)